

Principal Component Analysis (PCA)

Unsupervised Machine Learning

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How to Use This Document

Before You Start

- **Have a calculator ready** — We'll do lots of calculations together.
- **Work through every step** — Don't skip the calculations.
- **Use paper** — Write out each step as you read.
- **Go slowly** — This material builds on itself.
- **If confused** — Go back and review the previous section.
- **Check your work** — Verify each calculation before moving on.

What Will You Learn?

By the end of this guide, you will be able to:

1. Understand what PCA does and why we need it
2. Calculate the mean of a dataset
3. Compute a covariance matrix step-by-step
4. Find eigenvalues and eigenvectors (with detailed examples)
5. Transform data using principal components
6. Reduce data dimensions while keeping important information

Don't worry — we'll solve everything with actual numbers, step by step!

1 Part I: What is PCA?

1.1 The Big Picture

Key Idea #1

Principal Component Analysis (PCA) is a technique that simplifies complex data by reducing the number of variables (features) while keeping the most important information.

Think of it like this: Imagine you're taking a photo of a 3D object. You're reducing 3 dimensions to 2 dimensions (the photo), but you try to choose the best angle so you don't lose important details.

1.2 Why Do We Need PCA?

Real-World Problem: Suppose you're analyzing student performance with 100 different test scores. That's 100 dimensions! It's:

- Hard to visualize
- Slow to process
- Many features might be related (redundant)

PCA Solution: Maybe those 100 scores can be summarized by just 2 or 3 "super-features" (principal components) that capture most of the variation.

1.3 Simple Analogy

Imagine a classroom:

- You measure each student's height and arm length
- These two measurements are highly related (tall people usually have long arms)
- PCA finds a new measurement like "overall size" that combines both
- Instead of tracking 2 numbers, you track 1 number that captures most of the information

Stop and Think

Quick Check:

If you have data with 5 features, and PCA reduces it to 2 features, what happened to the other 3?

Answer: The 2 principal components are combinations of all 5 original features. We're not deleting features; we're combining them in a smart way!

2 Part II: The PCA Algorithm (Overview)

2.1 The 5 Steps of PCA

PCA Recipe — 5 Steps

- Step 1:** Standardize the data (make all features comparable)
- Step 2:** Calculate the covariance matrix (see how features relate)
- Step 3:** Find eigenvalues of the covariance matrix
- Step 4:** Find eigenvectors of the covariance matrix
- Step 5:** Transform the data using principal components

Don't panic! We'll explain each step with a complete numerical example.

3 Part III: Complete Numerical Example

3.1 The Dataset

Let's work with a small, simple dataset. We have 2 features (X_1 and X_2) measured for 4 examples.

Our goal: Reduce from 2 dimensions (features) to 1 dimension using PCA.

Feature	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

Table 1: Our sample dataset

3.2 Step 1: Calculate the Mean

Step 1: Calculate Mean

What: Find the average value for each feature.

Why: We need to center the data around zero.

Formula:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Calculation for X_1 :

$$\overline{X_1} = \frac{4 + 8 + 13 + 7}{4} = \frac{32}{4} = 8$$

Calculation for X_2 :

$$\overline{X_2} = \frac{11 + 4 + 5 + 14}{4} = \frac{34}{4} = 8.5$$

Result of Step 1

Mean values:

- $\overline{X_1} = 8$
- $\overline{X_2} = 8.5$

3.3 Step 2: Calculate the Covariance Matrix

Step 2: Covariance Matrix

What: The covariance matrix shows how features vary together.

Why: PCA needs to know which directions have the most variation.

Formula for covariance between X_i and X_j :

$$\text{Cov}(X_i, X_j) = \frac{1}{n} \sum_{k=1}^n (x_{ik} - \overline{X_i})(x_{jk} - \overline{X_j})$$

where n is the number of examples.

Let's calculate each entry:

Calculating $\text{Cov}(X_1, X_1)$ — Variance of X_1

$$\begin{aligned}
\text{Cov}(X_1, X_1) &= \frac{1}{4}[(4-8)(4-8) + (8-8)(8-8) \\
&\quad + (13-8)(13-8) + (7-8)(7-8)] \\
&= \frac{1}{4}[(-4)(-4) + (0)(0) + (5)(5) + (-1)(-1)] \\
&= \frac{1}{4}[16 + 0 + 25 + 1] \\
&= \frac{42}{4} = 10.5
\end{aligned}$$

Let me show you the table for clarity:

Example	x_1	$x_1 - \bar{X}_1$	$(x_1 - \bar{X}_1)^2$
1	4	$4 - 8 = -4$	$(-4)^2 = 16$
2	8	$8 - 8 = 0$	$(0)^2 = 0$
3	13	$13 - 8 = 5$	$(5)^2 = 25$
4	7	$7 - 8 = -1$	$(-1)^2 = 1$
Sum:			42

So $\text{Cov}(X_1, X_1) = \frac{42}{4} = 10.5$

Calculating $\text{Cov}(X_2, X_2)$ — Variance of X_2

$$\begin{aligned}
\text{Cov}(X_2, X_2) &= \frac{1}{4}[(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2] \\
&= \frac{1}{4}[(2.5)^2 + (-4.5)^2 + (-3.5)^2 + (5.5)^2] \\
&= \frac{1}{4}[6.25 + 20.25 + 12.25 + 30.25] \\
&= \frac{69}{4} = 17.25
\end{aligned}$$

Calculating $\text{Cov}(X_1, X_2)$ — Covariance between X_1 and X_2

$$\begin{aligned}
\text{Cov}(X_1, X_2) &= \frac{1}{4}[(4-8)(11-8.5) + (8-8)(4-8.5) \\
&\quad + (13-8)(5-8.5) + (7-8)(14-8.5)] \\
&= \frac{1}{4}[(-4)(2.5) + (0)(-4.5) + (5)(-3.5) + (-1)(5.5)] \\
&= \frac{1}{4}[-10 + 0 - 17.5 - 5.5] \\
&= \frac{-33}{4} = -8.25
\end{aligned}$$

Note: $\text{Cov}(X_2, X_1) = \text{Cov}(X_1, X_2) = -8.25$ (covariance is symmetric)

The Covariance Matrix

Result of Step 2

The covariance matrix C is:

$$C = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix} = \begin{bmatrix} 10.5 & -8.25 \\ -8.25 & 17.25 \end{bmatrix}$$

Stop and Think

What does the covariance matrix tell us?

- Diagonal entries (10.5 and 17.25): How much each feature varies
- Off-diagonal entry (-8.25): How X_1 and X_2 vary together (negative means when one goes up, the other tends to go down)

3.4 Step 3: Find Eigenvalues

Step 3: Eigenvalues

What: Eigenvalues tell us how much variance is captured by each principal component.

How: Solve the characteristic equation: $\det(C - \lambda I) = 0$

where I is the identity matrix and λ is the eigenvalue.

Set up the equation:

$$C - \lambda I = \begin{bmatrix} 10.5 & -8.25 \\ -8.25 & 17.25 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 10.5 - \lambda & -8.25 \\ -8.25 & 17.25 - \lambda \end{bmatrix}$$

Calculate the determinant:

$$\begin{aligned} \det(C - \lambda I) &= (10.5 - \lambda)(17.25 - \lambda) - (-8.25)(-8.25) \\ &= 181.125 - 10.5\lambda - 17.25\lambda + \lambda^2 - 68.0625 \\ &= \lambda^2 - 27.75\lambda + 113.0625 \end{aligned}$$

Solve $\lambda^2 - 27.75\lambda + 113.0625 = 0$ using the quadratic formula:

$$\lambda = \frac{27.75 \pm \sqrt{27.75^2 - 4(1)(113.0625)}}{2}$$

$$\begin{aligned} \lambda &= \frac{27.75 \pm \sqrt{770.0625 - 452.25}}{2} \\ &= \frac{27.75 \pm \sqrt{317.8125}}{2} \\ &= \frac{27.75 \pm 17.83}{2} \end{aligned}$$

Two solutions:

$$\begin{aligned} \lambda_1 &= \frac{27.75 + 17.83}{2} = \frac{45.58}{2} \approx 22.79 \\ \lambda_2 &= \frac{27.75 - 17.83}{2} = \frac{9.92}{2} \approx 4.96 \end{aligned}$$

Result of Step 3

Eigenvalues:

- $\lambda_1 \approx 22.79$ (larger eigenvalue — more important)
- $\lambda_2 \approx 4.96$ (smaller eigenvalue — less important)

Interpretation: The first principal component captures most of the variance!

3.5 Step 4: Find Eigenvectors

Step 4: Eigenvectors

What: Eigenvectors give us the directions of the principal components.

How: For each eigenvalue λ , solve $(C - \lambda I)\mathbf{v} = \mathbf{0}$

Finding Eigenvector for $\lambda_1 = 22.79$

Substitute into the equation:

$$\begin{bmatrix} 10.5 - 22.79 & -8.25 \\ -8.25 & 17.25 - 22.79 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -12.29 & -8.25 \\ -8.25 & -5.54 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row:

$$-12.29v_1 - 8.25v_2 = 0 \implies v_2 = -\frac{12.29}{8.25}v_1 \approx -1.49v_1$$

Let $v_1 = 1$, then $v_2 \approx -1.49$.

So the eigenvector is: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1.49 \end{bmatrix}$

Normalize it (make its length equal to 1):

$$\text{Length} = \sqrt{1^2 + (-1.49)^2} = \sqrt{1 + 2.22} = \sqrt{3.22} \approx 1.79$$

$$\mathbf{u}_1 = \frac{1}{1.79} \begin{bmatrix} 1 \\ -1.49 \end{bmatrix} = \begin{bmatrix} 0.56 \\ -0.83 \end{bmatrix}$$

Finding Eigenvector for $\lambda_2 = 4.96$

Similarly, substitute $\lambda_2 = 4.96$:

$$\begin{bmatrix} 10.5 - 4.96 & -8.25 \\ -8.25 & 17.25 - 4.96 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5.54 & -8.25 \\ -8.25 & 12.29 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row:

$$5.54v_1 - 8.25v_2 = 0 \implies v_2 = \frac{5.54}{8.25}v_1 \approx 0.67v_1$$

Let $v_1 = 1$, then $v_2 \approx 0.67$.

So the eigenvector is: $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0.67 \end{bmatrix}$

Normalize it:

$$\text{Length} = \sqrt{1^2 + 0.67^2} = \sqrt{1 + 0.45} = \sqrt{1.45} \approx 1.20$$

$$\mathbf{u}_2 = \frac{1}{1.20} \begin{bmatrix} 1 \\ 0.67 \end{bmatrix} = \begin{bmatrix} 0.83 \\ 0.56 \end{bmatrix}$$

Result of Step 4

Normalized Eigenvectors (Principal Component Directions):

- First PC: $\mathbf{u}_1 = \begin{bmatrix} 0.56 \\ -0.83 \end{bmatrix}$ (most important direction)
- Second PC: $\mathbf{u}_2 = \begin{bmatrix} 0.83 \\ 0.56 \end{bmatrix}$ (less important direction)

3.6 Step 5: Transform the Data (Find Principal Components)

Step 5: Data Transformation

What: Project the original data onto the principal component directions.

How: For each data point, compute:

$$\text{PC}_1 = (x_1 - \overline{X_1}) \cdot u_{11} + (x_2 - \overline{X_2}) \cdot u_{12}$$

where u_{11} and u_{12} are the components of $\mathbf{u}_1$.

Let's transform each example:

Example 1: $X_1 = 4, X_2 = 11$

$$\begin{aligned} \text{PC}_1 &= (4 - 8)(0.56) + (11 - 8.5)(-0.83) \\ &= (-4)(0.56) + (2.5)(-0.83) \\ &= -2.24 - 2.08 \\ &= -4.32 \end{aligned}$$

Example 2: $X_1 = 8, X_2 = 4$

$$\begin{aligned} \text{PC}_1 &= (8 - 8)(0.56) + (4 - 8.5)(-0.83) \\ &= (0)(0.56) + (-4.5)(-0.83) \\ &= 0 + 3.74 \\ &= 3.74 \end{aligned}$$

Example 3: $X_1 = 13, X_2 = 5$

$$\begin{aligned} \text{PC}_1 &= (13 - 8)(0.56) + (5 - 8.5)(-0.83) \\ &= (5)(0.56) + (-3.5)(-0.83) \\ &= 2.80 + 2.91 \\ &= 5.71 \end{aligned}$$

Example 4: $X_1 = 7$, $X_2 = 14$

$$\begin{aligned} PC_1 &= (7 - 8)(0.56) + (14 - 8.5)(-0.83) \\ &= (-1)(0.56) + (5.5)(-0.83) \\ &= -0.56 - 4.57 \\ &= -5.13 \end{aligned}$$

3.7 Final Result

Result of Step 5 — Data Reduced to 1 Dimension!

Original Data (2 dimensions):

Transformed Data (1 dimension):

Success! We've reduced the data from 2 features to 1 feature while keeping the most important information (captured by the largest eigenvalue).

Stop and Think

How much information did we keep?

The percentage of variance explained by PC1:

$$\text{Percentage} = \frac{\lambda_1}{\lambda_1 + \lambda_2} \times 100\% = \frac{22.79}{22.79 + 4.96} \times 100\% \approx 82.1\%$$

So we kept about **82%** of the information using just 1 dimension instead of 2!

4 Part IV: Another Complete Example

Let's do one more example with different numbers to reinforce the concepts.

4.1 Dataset 2

Feature	Ex1	Ex2	Ex3	Ex4	Ex5	Ex6	Ex7	Ex8
X_1	1	3	3	5	5	6	8	9
X_2	2	3	5	4	6	5	7	8

Table 2: Dataset 2

I'll guide you through the first few steps. Try to complete the rest yourself!

4.2 Step 1: Calculate Mean

For X_1 :

$$\overline{X_1} = \frac{1 + 3 + 3 + 5 + 5 + 6 + 8 + 9}{8} = \frac{40}{8} = 5$$

For X_2 :

$$\overline{X_2} = \frac{2 + 3 + 5 + 4 + 6 + 5 + 7 + 8}{8} = \frac{40}{8} = 5$$

Ex	x_1	$x_1 - 5$	$(x_1 - 5)^2$
1	1	-4	16
2	3	-2	4
3	3	-2	4
4	5	0	0
5	5	0	0
6	6	1	1
7	8	3	9
8	9	4	16
Sum:			50

4.3 Step 2: Calculate Covariance Matrix (Partial)

For $\text{Cov}(X_1, X_1)$:

Create the deviation table:

$$\text{Cov}(X_1, X_1) = \frac{50}{8} = 6.25$$

Try calculating $\text{Cov}(X_2, X_2)$ and $\text{Cov}(X_1, X_2)$ yourself!

Hint: Follow the same process. The answers are approximately:

- $\text{Cov}(X_2, X_2) = 3.5$
- $\text{Cov}(X_1, X_2) = 4.25$

So the covariance matrix is:

$$C = \begin{bmatrix} 6.25 & 4.25 \\ 4.25 & 3.5 \end{bmatrix}$$

4.4 Practice Exercise

Now try the remaining steps yourself:

1. Find the eigenvalues by solving $\det(C - \lambda I) = 0$
2. Find the eigenvectors for each eigenvalue
3. Transform the data to find the principal components

Check your answers:

- $\lambda_1 \approx 9.43$, $\lambda_2 \approx 0.32$
- First PC captures about 97% of variance!

5 Part V: Understanding the Results

5.1 What Do Principal Components Mean?

Key Idea #2

Principal components are **new features** that are:

1. **Linear combinations** of the original features
2. **Uncorrelated** with each other (independent)
3. **Ordered by importance** (PC1 captures most variance, then PC2, etc.)

5.2 When to Use PCA

PCA is useful when:

- You have many features (high dimensionality)
- Features are correlated (redundant information)
- You want to visualize high-dimensional data
- You need to speed up machine learning algorithms
- You want to reduce noise in the data

PCA may not be ideal when:

- Features have very different meanings (like age and salary)
- You need to interpret individual features
- The data is not linearly related

5.3 How Many Principal Components to Keep?

Choosing Number of Components

Common approaches:

1. Percentage of variance explained:

- Keep enough PCs to explain 80-95% of total variance
- Calculate: $\frac{\sum \text{kept eigenvalues}}{\sum \text{all eigenvalues}} \times 100\%$

2. Scree plot:

- Plot eigenvalues in decreasing order
- Look for an "elbow" where eigenvalues drop sharply
- Keep components before the elbow

3. Kaiser criterion:

- Keep components with eigenvalues > 1

Example from our first dataset:

- $\lambda_1 = 22.79$, $\lambda_2 = 4.96$
- Total variance: $22.79 + 4.96 = 27.75$
- PC1 explains: $\frac{22.79}{27.75} \times 100\% = 82.1\%$
- Both PCs together: 100%

If we only keep PC1, we retain 82.1% of the information with half the dimensions!

6 Part VI: Practice Problems

6.1 Problem 1: Quick Calculations

Given this covariance matrix:

$$C = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

Tasks:

1. Find the eigenvalues
2. Which principal component is more important?
3. What percentage of variance does PC1 explain?

6.2 Problem 2: Conceptual Questions

Answer True or False:

1. PCA can increase the number of features in your dataset. ____
2. The first principal component always has the largest eigenvalue. ____
3. Principal components are always easier to interpret than original features. ____
4. PCA works well when features are highly correlated. ____
5. You must always keep all principal components. ____

6.3 Problem 3: Real-World Application

A dataset has 10 features. After PCA, you get these eigenvalues:

[15.2, 8.3, 3.1, 1.5, 0.8, 0.4, 0.3, 0.2, 0.1, 0.1]

Questions:

1. What is the total variance?
2. How many components explain at least 90% of variance?
3. If you keep only 3 components, what percentage do you retain?

6.4 Problem 4: Fill in the Blanks

Complete this summary:

"PCA reduces the _____ of data while keeping the most _____. It does this by finding new _____ called principal components. These are _____ combinations of the original features. The first principal component has the _____ eigenvalue and captures the _____ variance."

7 Part VII: Step-by-Step Summary

PCA Summary — The Complete Recipe

Input: Data with n features

Step 1: Calculate the mean of each feature

$$\overline{X}_i = \frac{1}{m} \sum_{j=1}^m x_{ij}$$

Step 2: Calculate the covariance matrix

$$\text{Cov}(X_i, X_j) = \frac{1}{m} \sum_{k=1}^m (x_{ik} - \overline{X}_i)(x_{jk} - \overline{X}_j)$$

Step 3: Find eigenvalues by solving

$$\det(C - \lambda I) = 0$$

Step 4: Find eigenvectors by solving

$$(C - \lambda I)\mathbf{v} = \mathbf{0}$$

Step 5: Transform data using

$$\text{PC}_i = \sum_{j=1}^n (x_j - \overline{X}_j) \cdot u_{ij}$$

Output: Data with fewer features (principal components)

8 Answers to Practice Problems

8.1 Problem 1 Answers

Characteristic equation:

$$\det \begin{bmatrix} 4 - \lambda & 2 \\ 2 & 3 - \lambda \end{bmatrix} = (4 - \lambda)(3 - \lambda) - 4 = \lambda^2 - 7\lambda + 8 = 0$$

Eigenvalues:

$$\lambda = \frac{7 \pm \sqrt{49 - 32}}{2} = \frac{7 \pm \sqrt{17}}{2}$$
$$\lambda_1 \approx 5.56, \quad \lambda_2 \approx 1.44$$

PC1 is more important (larger eigenvalue)

Percentage: $\frac{5.56}{5.56+1.44} \times 100\% \approx 79.4\%$

8.2 Problem 2 Answers

1. **False** — PCA reduces dimensions, not increases 2. **True** — By definition, PC1 has the largest eigenvalue 3. **False** — PCs are combinations, often harder to interpret 4. **True** — PCA works best with correlated features 5. **False** — You typically keep only the most important ones

8.3 Problem 3 Answers

Total variance: $15.2 + 8.3 + 3.1 + 1.5 + 0.8 + 0.4 + 0.3 + 0.2 + 0.1 + 0.1 = 30$

Cumulative percentages:

- PC1: $\frac{15.2}{30} = 50.7\%$
- PC1+PC2: $\frac{15.2+8.3}{30} = 78.3\%$
- PC1+PC2+PC3: $\frac{15.2+8.3+3.1}{30} = 88.7\%$
- PC1+PC2+PC3+PC4: $\frac{26.6+1.5}{30} = 93.7\% \leftarrow \text{exceeds } 90\%$

Answer: Need 4 components for 90% variance

With 3 components: $\frac{26.6}{30} \times 100\% = 88.7\%$

8.4 Problem 4 Answers

"PCA reduces the **dimensionality** of data while keeping the most **information/variance**. It does this by finding new **features** called principal components. These are **linear** combinations of the original features. The first principal component has the **largest** eigenvalue and captures the **most** variance."

Final Tips for Success

You Can Master PCA!

Remember these key points:

1. **Always calculate step-by-step** — Don't skip calculations
2. **Check your arithmetic** — Small errors compound
3. **Eigenvalues tell importance** — Larger = more important
4. **Eigenvectors give direction** — They define new axes
5. **Practice with numbers** — Do lots of examples
6. **Understand the goal** — Reduce dimensions, keep information

Most importantly: PCA seems complex, but it's just organized linear algebra. With practice, the pattern becomes clear!

Keep working through examples, and you'll succeed!